Chapter 16 — Spring Cloud and Microservices

A **microservice architecture** splits one application into small services that are deployed independently and talk over the network. That buys independent releases and scaling, and costs you every problem of **distributed systems** (systems whose parts run on different machines and can only talk over an unreliable network; see System Design Fundamentals): where is the other service, what if it is slow, how do clients find a single entry point, where does configuration live, how do you follow one request through ten services. **Spring Cloud** is a family of projects that gives Spring Boot applications ready-made answers to those problems.

This chapter is about the *Spring implementation*. The patterns themselves — why a gateway (one front door for all clients), how a circuit breaker (a switch that stops calls to a failing service) thinks, sagas (multi-service workflows made of local steps plus undo steps), CQRS (separate models for writing and for reading data) — are explained in the Microservices Patterns notes and the System Design Patterns modules; each section links to the right one instead of re-teaching it.

What you'll learn

1. Which Spring project solves which microservice problem, and whether you need Spring Cloud at all on Kubernetes.
2. The Spring Cloud **release train** that matches Boot 4.1 and how to import it.
3. **Config Server** and the config client: layered configuration, refresh at runtime (run).
4. **Service discovery**: Eureka, Consul, Kubernetes (explained, not run) and the "simple" static discovery (run).
5. **Client-side load balancing** with Spring Cloud LoadBalancer (run).
6. **Declarative clients**: OpenFeign vs Spring's HTTP interface clients (run).
7. **API gateway** with Spring Cloud Gateway (server Web MVC flavour): routes, predicates, filters, circuit-breaker fallback (run).
8. **Resilience**: Resilience4j circuit breaker, retry and rate limiter state transitions (run) vs Framework 7's `@Retryable` / `@ConcurrencyLimit`.
9. Distributed tracing across services, contract testing, messaging between services.

The code: four Spring Boot 4.1.1 projects on Java 25 in `code/` — `config-demo`, `service-clients`, `gateway-demo`, `resilience-demo` (7 tests, all run). Downstream services (the services a service calls) in the tests are tiny local HTTP servers on random ports, so nothing external is needed. Outputs are real `mvn -q verify` output (log noise trimmed as `...`).

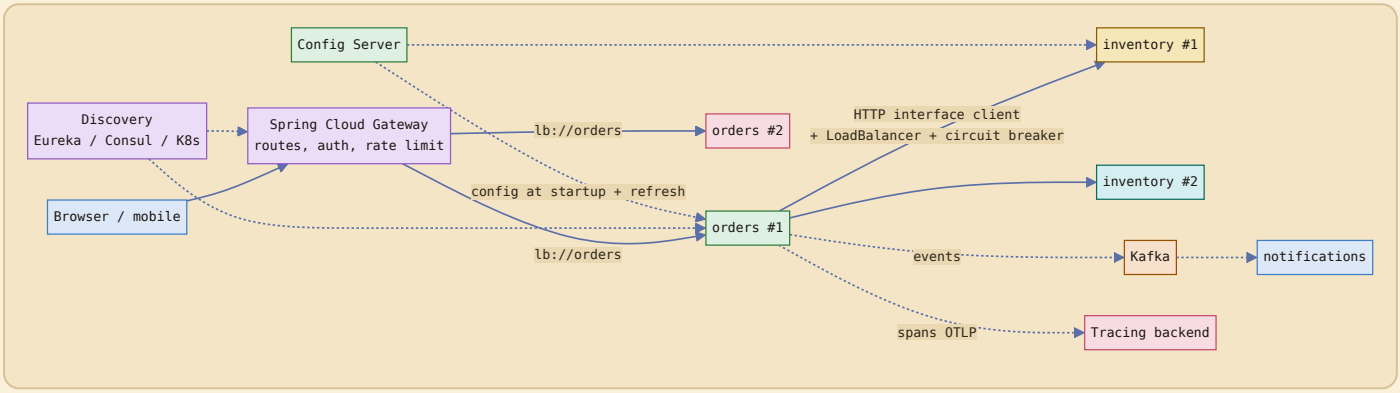
> **Prerequisites.** HTTP clients and HTTP interfaces: Chapter 6. Proxies and Framework 7 `@Retryable`: Chapter 4. Configuration and `spring.config.import`: Chapter 3. Tracing and health probes: Chapter 15. Messaging: Chapter 14.

16.1 The map: microservice problems and Spring answers

Why it matters

Before adding any dependency, know which problem it solves — and whether your platform already solves it. Many teams on Kubernetes (the standard platform for running containers across machines) still add Eureka and a Config Server out of habit, doubling the moving parts.

Problem	Pattern notes	Spring answer	On Kubernetes, often instead
One entry point for clients, cross-cutting edge concerns (things every request needs at the front door, such as auth and rate limits)	API Gateway, BFF	**Spring Cloud Gateway**	Ingress / Gateway API controller (Kubernetes' built-in HTTP entry points), service mesh (a network layer of sidecar proxies that handles routing, retries and encryption between services)
Find instances whose addresses change	Service Discovery	Eureka, Consul, Zookeeper, Spring Cloud Kubernetes	Kubernetes Services + DNS (the system that turns names into IP addresses)
Spread calls over instances	Load balancing	**Spring Cloud LoadBalancer** (client side)	kube-proxy (the Kubernetes component that forwards a Service's traffic to its pods) / mesh (server side)
Configuration per environment, changed without rebuild	Config externalization	**Spring Cloud Config**, Vault, Consul KV	ConfigMaps / Secrets (Kubernetes objects that hold configuration and secret values) (`spring.config.import=configtree:`)
Failing or slow dependencies	Circuit Breaker, Retry, Bulkhead	**Resilience4j** (+ Spring Cloud CircuitBreaker), Framework 7 `@Retryable` / `@ConcurrencyLimit`	mesh retries/outlier detection (automatically ejecting instances that keep failing) (coarser)
Following a request across services	Distributed Tracing	Micrometer Tracing + OpenTelemetry (15.7)	same, plus mesh telemetry
Declarative HTTP calls	—	HTTP interface clients (Framework), OpenFeign (maintenance)	—
Async communication, events	Event-Driven, Saga, CQRS	Spring Kafka / AMQP (Chapter 14), Spring Cloud Stream	—
Cross-cutting helpers next to each service	Sidecar	—	sidecar containers, service mesh



When not to use microservices

Distributed systems are harder to build, test, debug and operate. A small team, an unclear domain (the business area the software models) or no need for independent scaling are strong reasons to start with a **well-modularised monolith** (one deployable application with clean internal module boundaries) (Spring Modulith helps enforce module boundaries) and extract services later with the Strangler Fig approach. Also see DDIA (the book *Designing Data-Intensive Applications*) for the consistency problems you inherit.

16.2 Release trains and versions

What it is

Spring Cloud is ~15 separate projects (Commons, Config, Gateway, OpenFeign, Netflix, Consul, Kubernetes, Circuit Breaker, Stream, Function, Bus, Vault, Task, Zookeeper...), each with its own version. A **release train** is a BOM (bill of materials) that pins a set of versions known to work together and with one Spring Boot generation. Trains are named `YYYY.MINOR.PATCH` (since 2020) with a codename (a nickname such as "Oakwood").

Train	Codename	Spring Boot	Component versions
2025.1.x	**Oakwood**	**4.0.x, 4.1.x** (4.1 from 2025.1.2)	5.0.x
2025.0.x	Northfields	3.5.x	4.3.x — open-source support ended June 2026
2024.0.x	Moorgate	3.4.x	4.2.x

(Source: the compatibility table on spring.io/projects/spring-cloud; 2025.1.3, August 2026, is the latest service release at the time of writing and states it is based on Boot 4.0.x and compatible with 4.1.x. A `2026.0.0-M1` milestone (an early preview build) exists on Maven Central, the public repository Maven downloads libraries from.)

```
<properties>
  <spring-cloud.version>2025.1.3</spring-cloud.version>
</properties>
<dependencyManagement>
  <dependencies>
    <dependency>
      <groupId>org.springframework.cloud</groupId>
      <artifactId>spring-cloud-dependencies</artifactId>
      <version>${spring-cloud.version}</version>
      <type>pom</type>
      <scope>import</scope>
    </dependency>
  </dependencies>
</dependencyManagement>
<!-- then Spring Cloud starters without <version> -->
```

Things that changed recently and trip people up:

- **Gateway artifacts were renamed** (2025.0): `spring-cloud-starter-gateway-server-webflux` (reactive) and `spring-cloud-starter-gateway-server-webmvc` (servlet); properties moved to `spring.cloud.gateway.server.webflux.*` / `...server.webmvc.*`.
- **Spring Cloud Sleuth is gone** (last in 2021.0 for Boot 2.x); tracing is Micrometer Tracing in Boot itself.
- **Netflix Hystrix, Ribbon, Zuul** were removed years ago; use Resilience4j, Spring Cloud LoadBalancer and Spring Cloud Gateway.

- **OpenFeign** is feature-complete (maintenance mode) — the Spring team recommends HTTP interface clients.
- **Spring Cloud Contract**: the 2025.1.0–2025.1.2 BOMs manage it (5.0.x), but the 2025.1.3 BOM no longer lists it, and its GitHub repository has been moved to the `spring-attic` organisation and archived (read-only; last commit July 2026). Existing users can pin `spring-cloud-contract-dependencies` 5.0.3 explicitly; for new projects consider consumer-driven contracts with Pact (16.10).
- Resilience4j is **not** managed by Spring Boot; the Spring Cloud CircuitBreaker BOM pins Resilience4j 2.3.0, while Resilience4j's own Boot 4 module is `resilience4j-spring-boot4` 2.4.0 (used standalone in 16.8).

Common mistakes

- Mixing trains and Boot generations (e.g. 2025.0 with Boot 4) → `NoSuchMethodError` / `ClassNotFoundException` at startup. Spring Cloud checks compatibility on startup and fails fast (stops immediately instead of misbehaving later) with a clear message (`spring.cloud.compatibility-verifier.enabled`).
- Hard-coding component versions instead of importing the BOM.

16.3 Centralized configuration: Spring Cloud Config

What it is

A **Config Server** is a small Spring Boot app that serves configuration files over HTTP from a backend — usually a **Git repository** (a folder whose full change history is tracked by the Git version-control tool), also a local folder (`native` profile), Vault (HashiCorp's secret store), JDBC, S3 (Amazon's file storage) and others. Every service is a **config client**: at startup it imports its properties from the server, layered by *application name*, *profile* and *label* (Git branch/tag).

Analogy: a company handbook kept in one shared binder. Each department (service) reads the general chapter plus its own chapter; when the binder is updated, departments can re-read it without re-hiring staff.

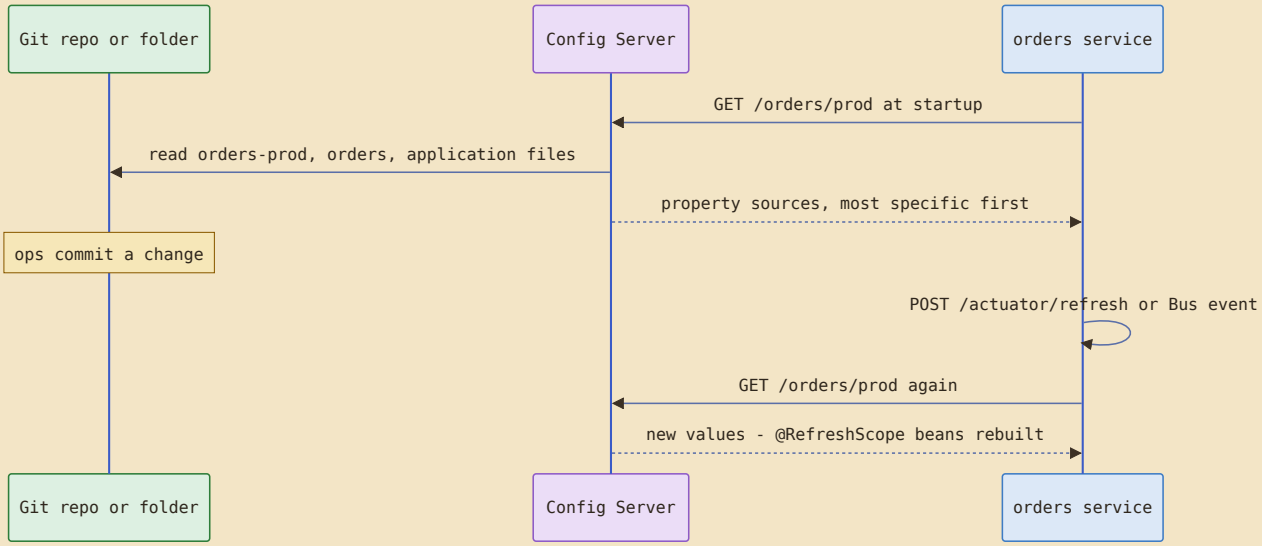
Why it matters

With 30 services and 4 environments you have 120 configuration sets. A Config Server gives one place to review, version (Git history = audit log, a record of who changed what and when), and change them — and to refresh running services without a rebuild. See Config externalization notes.

How it works

1. The server exposes `GET /{application}/{profile}[/{label}]` (plus file-style URLs like `/{application}-{profile}.properties`).
2. For `orders` + profile `prod` it returns an ordered list of property sources — most specific first: `orders-prod.properties` → `orders.properties` → `application-prod.properties` → `application.properties`.
3. The client adds `spring.config.import=configserver:http://config:8888` (Boot's config-data API; the old `bootstrap.yml` mechanism is legacy). Use `optional:configserver:...` if the service may start without it.
4. Remote properties are added to the `Environment` and **override the service's own `application.properties`**. Whether local system properties or environment variables (named values set outside the program by the OS or platform) may override remote values is controlled from the repository side (`spring.cloud.config.allow-override`, `override-none`, `override-system-properties`).

5. On refresh the client refetches, rebinds `@ConfigurationProperties` beans and re-creates `@RefreshScope` beans.



The demo

`ConfigServerAndClientTests` writes three files into a temporary folder (standing in for the Git repo), starts a Config Server with the `native` backend on a random port, then starts an `orders` client with profile `prod`:

```
// server
@SpringBootApplication
@EnableConfigServer
public class ConfigServerApplication { ... }
//   --spring.profiles.active=native
//   --spring.cloud.config.server.native.search-locations=file:<repo>/

// client bean
@Component
@RefreshScope   // re-created with fresh values after a refresh
public class OrderSettings {
    public OrderSettings(@Value("${orders.discount-percent}") int discountPercent,
                         @Value("${orders.greeting}") String greeting,
                         @Value("${shared.timeout}") String timeout) { ... }
}
//   --spring.application.name=orders --spring.profiles.active=prod
//   --spring.config.import=configserver:http://localhost:<port>
```

File in the repo	Content
`application.properties` (all services)	`shared.timeout=5s`, `orders.greeting=hello from application.properties`
`orders.properties`	`orders.greeting=hello from orders.properties`, `orders.discount-percent=10`
`orders-prod.properties`	`orders.discount-percent=15`

Then it edits `orders-prod.properties` to 20 and calls `ContextRefresher.refresh()` — what `POST /actuator/refresh` does:

```
GET /orders/prod -> property sources, highest priority first:
   orders-prod.properties
   orders.properties
   application.properties
client sees:    discount=15%, greeting='hello from orders.properties', timeout=5s
refresh changed [orders.discount-percent]
after refresh:  discount=20%, greeting='hello from orders.properties', timeout=5s
```

- `discount=15%` comes from the profile file, `greeting` from the service file (overriding the shared one), `timeout` from the shared file.
- `refresh()` returns exactly the keys that changed.

One demo-only detail: because server and client share one classpath here, the client passes `spring.cloud.config.enabled=true` — the server jar switches the client off by default (its `ConfigServerBootstrapApplicationListener`) so a Config Server does not try to fetch config from itself. In real deployments they are separate apps.

Refreshing many instances, secrets and alternatives

- **Spring Cloud Bus** (`spring-cloud-starter-bus-kafka` / `-amqp`) broadcasts one refresh to all instances (`POST /actuator/busrefresh`); the Config Server's `spring-cloud-config-monitor` can trigger it from a Git webhook (an HTTP call the Git host makes to your server when something is pushed).
- Values like `{cipher}AQB...` are decrypted by the server (`encrypt.key` or a keystore, a protected file that holds keys); better, keep secrets in **Vault** (`spring-cloud-starter-vault-config`) or Kubernetes Secrets.
- On Kubernetes, mounting ConfigMaps/Secrets as files and importing them with `spring.config.import=configtree:/etc/config/` needs no extra server (3.8); Spring Cloud Kubernetes can also read ConfigMaps via the API and reload.

Trade-offs and when not to use it

Pro	Con
One reviewed, versioned source; environment diffs visible	Another critical service: if it is down, new instances can't start (use `optional:`, `fail-fast` + retry, run ≥2 replicas, i.e. copies)
Runtime refresh without redeploy	Runtime refresh is also runtime *risk* — a bad value hits all instances at once
Works on any platform	On Kubernetes, ConfigMaps may be enough

Common mistakes

- Expecting every bean to see new values after refresh — only `@ConfigurationProperties` beans (rebound) and `@RefreshScope` beans (re-created) do. A `@Value` field in a normal singleton (a bean created once and shared, Spring's default) keeps the old value.
- Putting secrets in plain text in the Git repo.
- Forgetting `spring.application.name` — the client then asks for `application`, not `orders`.

Self-check

1. What does the server return for `/orders/prod`? → Property sources ordered most specific first: `orders-prod`, `orders`, (`application-prod`), `application`.
2. How does a Boot 4 client connect? → `spring.config.import=configserver:<url>` (optionally `optional:`).
3. Which beans pick up refreshed values? → `@ConfigurationProperties` beans and `@RefreshScope` beans.

16.4 Service discovery: Eureka, Consul, Kubernetes

What it is

In a dynamic environment instances come and go and their IP addresses (the numeric network addresses of machines) change. **Service discovery** keeps a **registry** of live instances per service name; clients ask "where is `inventory`?" instead of hard-coding addresses. Details and client-side vs server-side discovery: Service Discovery notes.

Spring Cloud Commons defines the abstraction — `DiscoveryClient` (`getInstances("inventory")` → list of `ServiceInstance` with host, port, metadata) — and each backend implements it:

Backend	Starter	How instances register	Health	Notes (not run here)
Netflix Eureka	`spring-cloud-starter-netflix-eureka-server` / `-client`	client registers itself and sends heartbeats ("I'm still alive" messages, every 30 s by default)	heartbeat lease expiry (dropped if heartbeats stop)	AP design (in CAP terms: stays **A**vailable during a network **P**artition rather than **C**onsistent): may serve stale (out-of-date) entries (self-preservation mode)
HashiCorp Consul	`spring-cloud-starter-consul-discovery`	client registers via the local Consul agent	Consul calls the app's health URL	also KV store for config (`spring-cloud-starter-consul-config`)
Kubernetes	`spring-cloud-starter-kubernetes-client-all` (or `-fabric8`)	Kubernetes itself (Endpoints of a Service)	readiness probes	needs RBAC (role-based access control) permission to read endpoints
Simple (in Commons)	none	static list in properties	none	tests, local dev, fixed topologies — used in 16.5
Zookeeper	`spring-cloud-starter-zookeeper-discovery`	ephemeral nodes (entries that vanish when their owner disconnects)	session	CP design (stays consistent, may refuse requests during a partition)

Eureka in a nutshell (not run):

```
@SpringBootApplication
@EnableEurekaServer                // the registry: a dashboard at http://localhost:8761
public class RegistryApplication { ... }
```

```
# each service (client) - with spring-cloud-starter-netflix-eureka-client on the classpath
spring.application.name: inventory                    # the service id
eureka:
  client:
    service-url:
      defaultZone: http://registry:8761/eureka/
  instance:
    prefer-ip-address: true
```

On Kubernetes, do you need it?

Usually **no**. A Kubernetes `Service` named `inventory` already gives a stable DNS name (`http://inventory:8080`) and kube-proxy spreads connections over ready pods (server-side discovery

and load balancing). Add Spring Cloud Kubernetes only if you need client-side features: instance metadata, per-request load balancing over pods (Kubernetes' running app instances), or discovery across namespaces (separate named areas inside one cluster). The table in 16.1 shows the platform equivalents.

Self-check

1. What does `DiscoveryClient.getInstances("inventory")` return? → The currently known `ServiceInstance`s (host, port, secure flag, metadata) for that service id.
2. Why is Eureka called AP? → During network partitions it keeps serving possibly stale registrations rather than refusing — availability over consistency.
3. Why might you skip Eureka on Kubernetes? → Services + DNS + readiness probes already provide registration, health and load balancing.

16.5 Client-side load balancing with Spring Cloud LoadBalancer

What it is

Client-side load balancing (load balancing = spreading requests over several instances; see System Design Fundamentals): the *caller* picks which instance of a service to call, from the list discovery provides — no load-balancer box in between. Spring Cloud LoadBalancer (the replacement for Netflix Ribbon) resolves a service id in a URL (`http://inventory/...` or `lb://inventory`) to a concrete instance per request.

How it works

1. A client marked for load balancing (`@LoadBalanced RestClient.Builder`, a Feign client with only a `name`, an HTTP service group without a `base-url`, the gateway's `lb://` URIs) makes a request to `http://inventory/stock/1`.
2. An interceptor asks the `ReactorLoadBalancer` for service `inventory`.
3. A `ServiceInstanceListSupplier` gets instances from the `DiscoveryClient` (cached), optionally filtered (health-check, zone, hints).
4. The algorithm picks one — **round robin** by default (take the instances in turn: A, B, A, B; start position randomised), or `RandomLoadBalancer`, weighted, "same instance preference", subset.
5. The URL is rewritten to `http://host:port/stock/1` and sent.

The demo: static instances, two clients

`service-clients` uses the **simple** discovery client — instances listed in properties — so nothing else must run. The test starts two tiny HTTP servers, "A" and "B", and points the properties at them:

```
spring.application.name=orders
# "Simple" discovery: a fixed list of instances per service id (instead of Eureka/Consul/Kubernetes).
spring.cloud.discovery.client.simple.instances.inventory[0].uri=http://localhost:9001
spring.cloud.discovery.client.simple.instances.inventory[1].uri=http://localhost:9002
# No spring.http.serviceclient.inventory.base-url: Spring Cloud LoadBalancer then uses
# the group name "inventory" as the service id and picks an instance per request.
```

Output (both clients shown in 16.6):

```
HTTP interface 4 calls served by [A, B, A, B]
OpenFeign      4 calls served by [A, B, A, B]  5th call -> Stock[sku=SKU-9, quantity=7, servedBy=A]
```

Round robin alternates between the instances. The starting instance is random, so another run may print `[B, A, B, A]` — the test asserts the alternation, not the order. The startup log also warns:

```
WARN ... LoadBalancerCaffeineWarnLogger : Spring Cloud LoadBalancer is currently working with the default cache. While this cache implementation is useful for development and tests, it's recommended to use Caffeine cache in pr...
```

— in production add `com.github.ben-manes.caffeine:caffeine` (Caffeine, a fast in-memory cache library) so the instance list cache is bounded and refreshed properly (`spring.cloud.loadbalancer.cache.ttl` — TTL = time to live, how long an entry is kept; default 35 s).

Useful settings:

```
# instance-list supplier chain: default | health-check | zone-preference | weighted | same-instance-preference | request-based-sticky-session | subset
spring.cloud.loadbalancer.configurations=health-check
# per-service overrides live under spring.cloud.loadbalancer.clients.<serviceId>.*
```

For a plain `RestClient`, mark the builder:

```
@Bean @LoadBalanced
RestClient.Builder loadBalancedBuilder() { return RestClient.builder(); }
// restClientBuilder.baseUrl("http://inventory").build()
```

Trade-offs

Client-side (Spring Cloud LoadBalancer)	Server-side (K8s Service, NGINX, cloud LB, mesh)
per-request choice, custom algorithms, zone/metadata aware	nothing in the app; works for every language
each client needs the library and a discovery source	extra hop (one more network step between caller and service; except kube-proxy/mesh sidecars), fewer app-level decisions

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16.6 Declarative clients: OpenFeign vs HTTP interface clients

What it is

Both let you declare a Java interface and get a generated proxy (an object Spring creates at runtime that implements the interface) that makes the HTTP calls (6.6 teaches HTTP interfaces in depth).

```java
// Spring Cloud OpenFeign: name = service id -> resolved by the LoadBalancer (no url given)
@FeignClient(name = "inventory")
public interface InventoryFeignClient {
    @GetMapping("/stock/{sku}")                        // Spring MVC annotations
    Stock stock(@PathVariable("sku") String sku);
}

// Spring Framework HTTP service interface (no extra library)
public interface InventoryHttpClient {
    @GetExchange("/stock/{sku}")
    Stock stock(@PathVariable String sku);
}

@SpringBootApplication
@EnableFeignClients                                                    // scan for @FeignClient
@ImportHttpServices(group = "inventory", types = InventoryHttpClient.class) // group name = service id
public class ClientsApplication { ... }
```

Since Spring Cloud 2025.1 (Commons 5.0), LoadBalancer plugs into Boot 4's **HTTP service client groups**: a group without `spring.http.serviceclient.<group>.base-url` gets `http://<groupName>` load-balanced, and a `base-url` with the `lb://` scheme is also supported. That's why the `inventory` group above needed no URL. The test (`LoadBalancedClientsTests`) calls each client four times — output in 16.5.

Comparison

	OpenFeign	**HTTP interface client**
Where it lives	Spring Cloud OpenFeign (separate project)	Spring Framework 6+ core; Boot 4 groups via `@ImportHttpServices`
Status	**feature-complete / maintenance** — only fixes	actively developed; the Spring team's recommendation
Annotations	Spring MVC's `@GetMapping`, ... on the interface	`@GetExchange`, `@PostExchange`, ...
Underlying client	Feign's own client (or Apache HC5 / OkHttp)	`RestClient` or `WebClient` (reactive return types possible)
Config	`spring.cloud.openfeign.client.config.<name>.*` (timeouts, loggers, interceptors)	`spring.http.serviceclient.<group>.*`, `RestClientHttpServiceGroupConfigurer`
Load balancing	automatic when only `name` is set	automatic for groups without base URL (Spring Cloud 2025.1+)
Circuit breaker	`spring.cloud.openfeign.circuitbreaker.enabled=true` + fallbacks	Spring Cloud 2025.1+: per-group breaker with an `@HttpServiceFallback` class (switch `spring.cloud.circuitbreaker.http-services.enabled`, on by default; not run here), or wrap the call in a service method with Resilience4j
Tracing	instrumented	instrumented (via the observed `RestClient`)

Recommendation: new code → HTTP interface clients; existing Feign code keeps working, migrate when convenient (the annotations map almost one-to-one).

16.7 API gateway: Spring Cloud Gateway

What it is

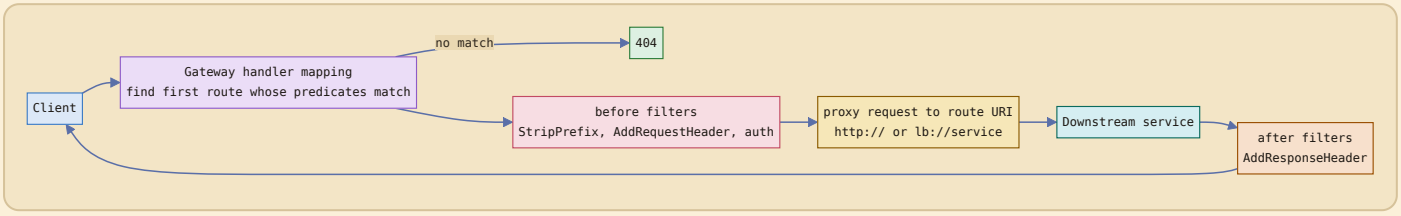
An **API gateway** is the single front door for external clients: it **routes** each request to the right service and applies **edge concerns** once — authentication/token relay, rate limiting, CORS (Cross-Origin Resource Sharing: the headers that tell browsers which other websites may call the API), header rewriting, retries/circuit breaking, request size limits. Pattern details and trade-offs: API Gateway notes, BFF, API & Edge module.

Spring Cloud Gateway comes in two flavours with the same concepts:

	Server Web MVC	**Server WebFlux**
Starter	`spring-cloud-starter-gateway-server-webmvc`	`spring-cloud-starter-gateway-server-webflux`
Runtime	Servlet (Tomcat/Jetty), blocking; pairs well with virtual threads	Netty, reactive
Routes in code	`RouterFunction` DSL (`GatewayRouterFunctions.route`)	`RouteLocatorBuilder` DSL
Properties	`spring.cloud.gateway.server.webmvc.routes`	`spring.cloud.gateway.server.webflux.routes`
Choose when	team knows MVC, uses blocking libraries	very high connection counts, streaming, already reactive

How it works

A **route** = id + destination URI + **predicates** (when does it match? `Path`, `Method`, `Header`, `Host`, `Query`, `Cookie`, `After`, `Weight`, `Version`...) + **filters** (what to change? `StripPrefix`, `AddRequestHeader`, `RewritePath`, `CircuitBreaker`, `Retry`, `RequestRateLimiter`, `TokenRelay`, ...).



The demo

`gateway-demo` defines one route in YAML and one in Java.

```yaml
spring:
  cloud:
    gateway:
      server:
        webmvc:
          routes:
            # Route 1, in YAML: /api/orders/** -> orders service
            - id: orders
              uri: ${services.orders.url}              # with discovery: lb://orders
              predicates:
                - Path=/api/orders/**
                - Method=GET,POST
              filters:
                - StripPrefix=1                          # /api/orders/7 -> /orders/7
                - AddRequestHeader=X-Gateway, ch16-gateway
                - AddResponseHeader=X-Served-Via, gateway
```

```java
/** Route 2, in the Java DSL: /api/inventory/** -> inventory service, protected by a circuit breaker.
*/
@Bean
RouterFunction<ServerResponse> inventoryRoute(@Value("${services.inventory.url}") String inventoryUrl)
{
    return route("inventory")
            .GET("/api/inventory/**", http())
            .before(uri(inventoryUrl))
            .filter(circuitBreaker("inventoryCB", URI.create("forward:/fallback/inventory")))
            .build();
}

@GetMapping("/fallback/inventory")
String inventoryFallback() {
    return "{\"items\":[],\"note\":\"inventory temporarily unavailable\"}";
}
```

(Static imports: `GatewayRouterFunctions.route`, `HandlerFunctions.http`, `BeforeFilterFunctions.uri`, `CircuitBreakerFilterFunctions.circuitBreaker`. Pass the target with `before(uri(...))`: the older `http(String)` / `http(URI)` overloads, deprecated in Gateway 4.x, no longer exist in Gateway 5.0 — only `http()` remains.) The `CircuitBreaker` filter uses Spring Cloud CircuitBreaker, so the project adds `spring-cloud-starter-circuitbreaker-resilience4j`.

`GatewayRoutingTests` runs an "orders service" that echoes what it received, and points the inventory route at a port where nothing listens:

```
GET /api/orders/7 -> orders-service saw GET /orders/7?expand=items X-Gateway=ch16-gateway | response
X-Served-Via=gateway
DELETE /api/orders/7 and GET /api/unknown -> 404 (no route matches)
GET /api/inventory/items (service down) -> {"items":[],"note":"inventory temporarily unavailable"}
```

- `StripPrefix=1` removed `/api`; the query string was kept; the request and response headers were added.
- `DELETE` fails the `Method` predicate and nothing matches `/api/unknown` → 404.
- The connection failure to inventory went through the circuit breaker to the fallback — the client got a degraded but valid answer.

Other common gateway jobs (not run)

- **Token relay** (`TokenRelay` filter with `spring-boot-starter-oauth2-client`): the gateway logs the user in and forwards the access token to services — see 11.10 and the BFF discussion in 11.12.
- **Rate limiting**: WebFlux flavour has `RequestRateLimiter` backed by Redis; the MVC flavour offers a Bucket4j-based filter. See Rate Limiting.
- **Weighted routes** (`Weight=group, 90`) for canary releases (sending a small share of traffic to a new version first).

Common mistakes

- Putting business logic or aggregation of many services in the gateway → it becomes a "distributed monolith" bottleneck. Keep it thin; use a BFF for client-specific aggregation.
- Using Spring Cloud Gateway *and* an Ingress/cloud gateway that do the same things — pick one layer for each concern.
- Retrying non-idempotent `POST`s at the gateway (an *idempotent* call can safely run twice with the same effect; a `POST` that creates an order usually cannot).

16.8 Resilience: circuit breaker, retry, rate limiter

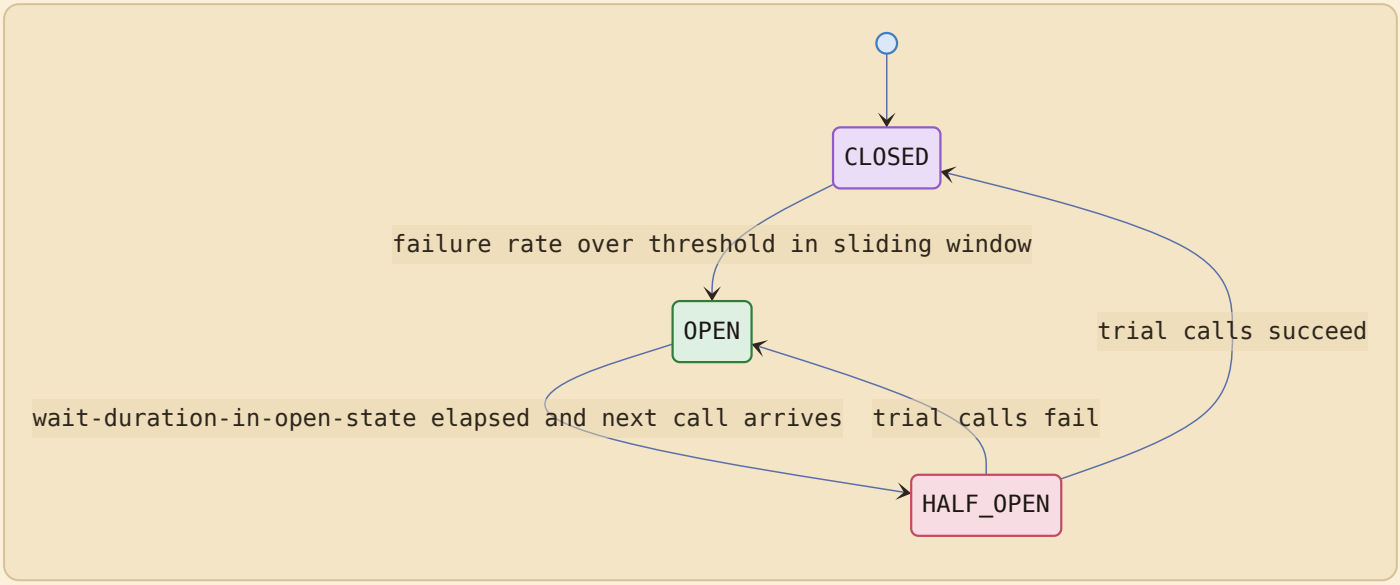
What it is

Remote calls fail and, worse, become **slow**. Without protection, callers pile up waiting threads until they fail too — a **cascading failure**. The main tools (Reliability module):

Tool	Idea	Notes
Timeout	never wait forever	the first line of defence; set on every client (Chapter 6)
Retry	try again after a transient (temporary) failure, with back-off (growing waits) + jitter (a random extra wait so clients don't retry in lockstep)	only idempotent operations — Retry notes
Circuit breaker	after many failures, stop calling for a while and fail fast / fall back	Circuit Breaker notes
Bulkhead	cap concurrent calls to one dependency	Bulkhead notes
Rate limiter	cap calls per time period	protects a dependency or enforces a quota (an agreed maximum usage)
Fallback	a degraded answer (cache, default, empty list)	Graceful degradation

Resilience4j is the standard Java library for all of these (Hystrix's successor). Use it directly (`resilience4j-spring-boot4`: annotations + properties + Micrometer metrics + health) or through **Spring Cloud CircuitBreaker**, an abstraction (`CircuitBreakerFactory.create("id").run(supplier, fallback)`) with a Resilience4j implementation, used e.g. by the gateway filter above.

Circuit breaker states



- **CLOSED** — calls pass; outcomes are recorded in a **sliding window** (last N calls, or last N seconds).
- **OPEN** — calls are rejected immediately with `CallNotPermittedException` (→ fallback) without touching the dependency, giving it time to recover.
- **HALF_OPEN** — a limited number of trial calls decide whether to close again.

The demo

`resilience-demo` wraps a fake remote `RemoteInventory` (the tests switch it up or down):

```java
@CircuitBreaker(name = "inventory", fallbackMethod = "cachedStock")
public String stockWithBreaker(String sku) {
    return "live stock " + remote.stock(sku);
}

// fallback: same parameters + the exception as the last parameter
private String cachedStock(String sku, Throwable t) {
    return "cached stock 5 (" + t.getClass().getSimpleName() + ")";
}

@Retry(name = "inventory-retry")
public String stockWithRetry(String sku) { return "live stock " + remote.stock(sku); }

@RateLimiter(name = "search")
public String search(String text) { return "results for " + text; }
```

```yaml
resilience4j:
  circuitbreaker:
    instances:
      inventory:
        sliding-window-type: COUNT_BASED
        sliding-window-size: 4                # judge the last 4 calls
        minimum-number-of-calls: 4            # ... but only once 4 calls were recorded
        failure-rate-threshold: 50            # open when >= 50 % of them failed
        wait-duration-in-open-state: 300ms    # stay OPEN this long, then allow trial calls
        permitted-number-of-calls-in-half-open-state: 2
        record-exceptions: ch16.resilience.RemoteCallException
  retry:
    instances:
      inventory-retry:
        max-attempts: 3                       # total attempts, including the first call
        wait-duration: 20ms
        enable-exponential-backoff: true
        exponential-backoff-multiplier: 2     # 20 ms, then 40 ms
        retry-exceptions: ch16.resilience.RemoteCallException
  ratelimiter:
    instances:
      search:
        limit-for-period: 3                   # 3 permits ...
        limit-refresh-period: 10s             # ... per 10-second cycle
        timeout-duration: 0                   # don't wait for a permit: reject immediately
```

`CircuitBreakerTests` subscribes to the breaker's state-transition events and walks it through all states:

```
1) dependency goes down, 4 calls:
   call 1: cached stock 5 (RemoteCallException)  state=CLOSED
   call 2: cached stock 5 (RemoteCallException)  state=CLOSED
   call 3: cached stock 5 (RemoteCallException)  state=CLOSED
   call 4: cached stock 5 (RemoteCallException)  state=OPEN
2) while OPEN the remote is not even called:
   call 5: cached stock 5 (CallNotPermittedException)  remote calls +0
3) wait 300 ms, dependency is back; 2 trial calls in HALF_OPEN:
   call 6: live stock 7  state=HALF_OPEN
   call 7: live stock 7  state=CLOSED
transitions: [CLOSED -> OPEN, OPEN -> HALF_OPEN, HALF_OPEN -> CLOSED]
```

- Calls 1–3 fail but the breaker stays CLOSED: the window needs `minimum-number-of-calls` (4) before judging.

- After call 4 the failure rate is 100% ≥ 50% → OPEN.
- Call 5 is rejected without calling the remote (`remote calls +0`) — this is what stops the cascade.
- After 300 ms the next call moves the breaker to HALF_OPEN — by default (`automatic-transition-from-open-to-half-open-enabled: false`) no timer does this; the first call after the wait triggers it. Two successful trials close it.
- Without these settings Resilience4j's defaults apply: `COUNT_BASED` window of 100 calls, `minimum-number-of-calls` 100, `failure-rate-threshold` 50, `wait-duration-in-open-state` 60 s, 10 permitted calls in HALF_OPEN — sensible for production traffic, far too slow for a demo. Retry defaults to 3 attempts 500 ms apart.

`RetryAndRateLimiterTests`:

```
Framework 7 @Retryable, remote fails twice -> live stock 7 after 3 attempts
RateLimiter (3 per 10 s): call1=OK call2=OK call3=OK call4=REJECTED call5=REJECTED
Resilience4j @Retry, remote fails twice:
   retry #1 after inventory unavailable, waiting 20 ms
   retry #2 after inventory unavailable, waiting 40 ms
   -> live stock 7 after 3 attempts
   retry #1 after inventory unavailable, waiting 20 ms
   retry #2 after inventory unavailable, waiting 40 ms
   remote down: gave up after 3 attempts, caller gets RemoteCallException
```

- Resilience4j's `max-attempts: 3` counts the **first call**; Framework 7's `maxRetries = 2` counts only **retries** — both mean 3 attempts.
- Exponential back-off doubled the wait (20 → 40 ms). Add jitter in production (`randomized-wait-factor` / `enable-randomized-wait`).
- The rate limiter rejects with `RequestNotPermitted` once the 3 permits of the current period are used.

The Framework 7 version used in the comparison (Chapter 4.7 covers it fully):

```
@Retryable(includes = RemoteCallException.class, maxRetries = 2, delay = 20, multiplier = 2)
@ConcurrencyLimit(limit = 10, policy = ConcurrencyLimit.ThrottlePolicy.REJECT)   // a simple bulkhead
public String stock(String sku) { return "live stock " + remote.stock(sku); }
// needs @EnableResilientMethods on a configuration class
```

Resilience4j vs Framework 7 core

Need	Framework 7 core	Resilience4j
Retry with back-off, jitter	`@Retryable`, `RetryTemplate`	`@Retry`, `Retry`
Bulkhead (concurrency cap)	`@ConcurrencyLimit` (BLOCK or REJECT)	`@Bulkhead` (semaphore — a counter of free slots — or thread pool)
Circuit breaker	—	`@CircuitBreaker`
Rate limiter	—	`@RateLimiter`
Time limiter	— (use client timeouts)	`@TimeLimiter` (async/reactive)
Fallback methods	— (catch at the caller)	`fallbackMethod`
Config in properties, per instance	attributes (placeholders allowed)	yes, rich
Metrics, health, events	`MethodRetryEvent`	Micrometer meters, health indicators, event publishers
Extra dependency	none	yes

Rule of thumb: plain retries and concurrency caps → Framework 7; anything involving circuit breaking, rate limiting or fallbacks → Resilience4j.

Order when stacking annotations (Resilience4j defaults): `Retry( CircuitBreaker( RateLimiter( TimeLimiter( Bulkhead( call ) ) ) ) )` — the retry is outermost, so each retry attempt is counted by the breaker, and an OPEN breaker's rejection can be retried unless you exclude `CallNotPermittedException`. Change the order with `resilience4j.*.<type>-aspect-order` properties.

Common mistakes

- Retrying everything, including 4xx errors and non-idempotent calls.
- Retries at gateway **and** client **and** service → retry storms (3 × 3 × 3 = 27 calls).
- A circuit breaker without timeouts: slow calls never "fail", so the breaker never opens (or configure `slow-call-duration-threshold` / `slow-call-rate-threshold`).
- One breaker for all endpoints of a service — one bad endpoint opens the circuit for the healthy ones.
- Self-invocation — all these annotations are proxy-based (Chapter 4).
- Fallbacks that hide outages forever — alert on breaker state (`resilience4j_circuitbreaker_state` metric).

Self-check

1. What happens to calls while a breaker is OPEN? → Rejected immediately with `CallNotPermittedException` (fallback if defined), without calling the dependency.
2. Why did the breaker stay CLOSED after 3 failures? → `minimum-number-of-calls` was 4.
3. Resilience4j `max-attempts: 3` vs Framework `maxRetries = 3`? → 3 total attempts vs 4 total attempts.
4. Which is the outermost Resilience4j aspect by default? → Retry.

TL;DR

Timeouts first; retry only transient, idempotent failures with back-off + jitter; circuit breaker to fail fast and give the dependency room; bulkhead/rate limiter to cap load; fallbacks for degraded answers. Framework 7 covers retry + concurrency limits; Resilience4j covers everything.

16.9 Distributed tracing across services

Little is needed beyond Chapter 15.7: every service adds `spring-boot-starter-opentelemetry` and exports to the same collector; the gateway and clients built from Boot's `RestClient.Builder` (including HTTP interface groups) and OpenFeign propagate the W3C `traceparent` header. For messaging, switch observation on explicitly (e.g. `spring.kafka.template.observation-enabled=true` and `spring.kafka.listener.observation-enabled=true`, same idea for RabbitMQ) so the context travels in message headers. Things specific to microservices:

- Use the **same propagation format** everywhere (W3C by default; set `management.tracing.propagation.type=b3` only for legacy Zipkin/B3 services, or produce both).
- Set `spring.application.name` in every service — it becomes the service name in traces and logs.
- Let the **edge decide sampling** (the gateway) — downstream services follow the sampled flag.
- Tag spans with business ids as **high-cardinality** values (order id), never as metric tags.

16.10 Contract testing

What it is

Service A (consumer) and service B (provider) are tested separately — how do you know they still agree on the API? End-to-end tests of all services are slow and flaky. **Contract testing** checks each side against a shared **contract** (request → expected response):

1. Consumer-driven: the consumer writes what it needs.
2. The provider's build **verifies** it really responds that way.
3. The consumer's tests run against a **stub** (a fake server that returns canned answers) generated from the same contract.

Tool	How	Status
Pact (pact-jvm)	consumer tests produce a pact file; a Pact Broker (a server that stores contracts and results) shares it; provider verifies; `can-i-deploy` gate	active, polyglot (works across programming languages)
Spring Cloud Contract	contracts in YAML/Groovy/Java in the provider repo; the plugin generates provider tests and a stubs jar; consumers use `@AutoConfigureStubRunner`	JVM/Spring-centric; not in the 2025.1.3 BOM, repository archived in `spring-attic` (see 16.2)
OpenAPI-based checks	validate both sides against the OpenAPI document	good for provider-driven public APIs

Spring Cloud Contract, for recognition (not run):

```yaml
# provider: src/test/resources/contracts/stock/shouldReturnStock.yml
request:
  method: GET
  url: /stock/SKU-1
response:
  status: 200
  headers: { Content-Type: application/json }
  body: { sku: "SKU-1", quantity: 7 }
```

```
// consumer test: runs against a WireMock stub built from the provider's contracts
@SpringBootTest
@AutoConfigureStubRunner(ids = "com.acme:inventory:+:stubs:8090", stubsMode =
StubRunnerProperties.StubsMode.LOCAL)
class InventoryClientContractTest { /* call http://localhost:8090/stock/SKU-1 */ }
```

When not to use it: a single team owning both sides that deploys them together, or public APIs where you can't know consumers (use versioning + OpenAPI instead).

16.11 Messaging between services

Synchronous HTTP chains couple availability: if any service in the chain is down, the request fails. Events decouple them in time (Event-Driven notes). Spring options:

- **Spring for Apache Kafka / Spring AMQP** — templates and `@KafkaListener` / `@RabbitListener` (Chapter 14, including the transactional outbox).
- **Spring Cloud Stream** — a binder abstraction: you write `java.util.function` beans (`Supplier`, `Function`, `Consumer`) and bind them to Kafka/RabbitMQ destinations by configuration (`spring.cloud.function.definition=enrichOrder`, `spring.cloud.stream.bindings.enrichOrder-in-0.destination=orders`). Good for many similar pipelines; it hides broker-specific features you may need.
- Cross-service consistency without distributed transactions (one transaction spanning several databases): Saga (choreography — each service reacts to others' events — or orchestration — one coordinator tells each service what to do), idempotent consumers, and CQRS read models.

16.12 Common mistakes (whole chapter)

Symptom	Cause	Fix
App fails at startup with a compatibility error	Spring Cloud train doesn't match Boot	2025.1.x for Boot 4.0/4.1
`No spring.config.import property has been defined`	config client on classpath without import	add `spring.config.import=configserver:...` (or `optional:`)
Config change not visible after refresh	plain `@Value` in a singleton	`@RefreshScope` or `@ConfigurationProperties`
`UnknownHostException: inventory`	client not load-balanced	`@LoadBalanced` builder, Feign `name`, or HTTP group without base URL
LoadBalancer "default cache" warning	Caffeine missing	add Caffeine
Gateway returns 404	predicates don't match (method, path, prefix)	check the route definition; DEBUG logging for `org.springframework.cloud.gateway`
Breaker never opens	no timeouts, slow calls not recorded, window too large	timeouts + slow-call thresholds + sensible window
Retry storm during an outage	retries at several layers	retry in one layer, with jitter; breaker around retries
Everything down when one service is down	synchronous chains, no fallbacks	timeouts, breakers, fallbacks, async events

---

16.13 In the interview

Q: Which Spring Cloud version do you use with Spring Boot 4.1? The 2025.1.x "Oakwood" train (Boot 4.0 and 4.1, the latter from 2025.1.2), imported as the `spring-cloud-dependencies` BOM; its components are 5.0.x.

Q: How do microservices find each other in Spring? Through the `DiscoveryClient` abstraction backed by Eureka, Consul, Zookeeper or Kubernetes; Spring Cloud LoadBalancer resolves service ids (`http://inventory`, `lb://inventory`) to instances per request, round robin by default. On Kubernetes plain Services + DNS usually suffice.

- *Follow-up: client-side vs server-side load balancing?* → Client picks the instance (smarter, per-request, needs a library) vs a proxy picks it (language-agnostic, extra hop).

Q: Explain the circuit breaker states and how you configure them in Resilience4j. CLOSED records outcomes in a sliding window; when the failure (or slow-call) rate exceeds the threshold after `minimum-number-of-calls`, it goes OPEN and rejects calls fast; after `wait-duration-in-open-state` it goes HALF_OPEN and lets `permitted-number-of-calls-in-half-open-state` trial calls decide between CLOSED and OPEN. Configure per instance under `resilience4j.circuitbreaker.instances.<name>`; use `@CircuitBreaker(name, fallbackMethod)`.

- *Follow-up: how is this different from retry?* → Retry repeats a single call hoping for success; the breaker stops calling at all when the dependency is clearly unhealthy. Combine them, with the breaker protecting against retry storms.
- *Follow-up: does Spring Framework 7 have a circuit breaker?* → No — it added `@Retryable` and `@ConcurrencyLimit` only.

Q: What does an API gateway do, and which Spring Cloud Gateway flavour would you pick? Single entry point: routing by predicates, filters for auth/token relay, rate limiting, header rewriting, resilience. Web MVC flavour for servlet teams and virtual threads; WebFlux flavour for reactive stacks and very high concurrency. Keep business logic out.

Q: How do you manage configuration across 30 services? Externalised config: Spring Cloud Config (Git-backed, layered by app/profile/label, refresh via actuator or Bus) or platform-native ConfigMaps/Secrets with `configtree:`; secrets in Vault; config changes reviewed like code.

Q: OpenFeign or HTTP interfaces? → HTTP interfaces for new code (core Spring, actively developed, Boot 4 groups, load-balanced by Spring Cloud); OpenFeign is in maintenance mode.

Q: How do you test that services stay compatible? → Consumer-driven contract tests (Pact, or Spring Cloud Contract in existing projects) plus a small number of end-to-end smoke tests.

Chapter TL;DR

- Spring Cloud = ready-made answers to distributed-system problems; check what your platform (Kubernetes) already gives you before adding them. Patterns: microservices-patterns notes.
- **Boot 4.1 → Spring Cloud 2025.1.x (Oakwood)**, BOM `spring-cloud-dependencies` (2025.1.3 used here); components 5.0.x; OpenFeign in maintenance; Spring Cloud Contract no longer in the 2025.1.3 BOM.
- **Config Server**: `@EnableConfigServer`, Git or `native` backend; clients use `spring.config.import=configserver:`; precedence `app-profile` > `app` > `application`; refresh re-creates `@RefreshScope` beans (proved: 15% → 20%).
- **Discovery**: `DiscoveryClient` over Eureka/Consul/Kubernetes/simple; **LoadBalancer** resolves service ids per request, round robin by default (proved: B, A, B, A); add Caffeine.
- **Declarative clients**: prefer HTTP interfaces (`@ImportHttpServices`, load-balanced when the group has no base URL); OpenFeign still works.
- **Gateway**: routes = predicates + filters; server-webmvc or server-webflux; `StripPrefix`, headers, `CircuitBreaker` with fallback (proved).
- **Resilience4j**: breaker CLOSED → OPEN → HALF_OPEN → CLOSED (proved), retry with back-off, rate limiter (proved); Framework 7 `@Retryable`/`@ConcurrencyLimit` for simple cases.
- Tracing across services = Chapter 15 setup in every service; contracts with Pact/Spring Cloud Contract; events with Kafka/Stream.
- Next: Chapter 17 — Spring AI and what's new.